## Introduction

In modern cities, population and vehicle growth have made parking management a major challenge. Traditional parking systems consume large spaces and cause congestion and fuel wastage. The vertical car parking system offers a smart, space-efficient solution by storing cars vertically using mechanical structures. Operated by electric motors, and control systems, it lifts and retrieves vehicles safely without human effort. This system increases parking capacity, saves time, and supports smarter urban infrastructure.

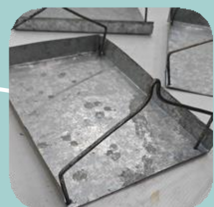
## Purpose of Vertical Rotatory Car Parking

The purpose of this project is to design a vertical rotary car parking system that efficiently utilizes vertical space to park multiple vehicles within a small area, reducing land usage, parking time, and human effort through a safe and reliable motorized mechanism.

## Working Procedure of Vertical Rotary Car Parking

1. Drive the vehicle onto the parking platform.
2. Activate the parking system.
3. The motor rotates the platforms vertically.
4. The sensor stops the platform at the required position.
5. For retrieval, the system rotates until the vehicle reaches the entrance.
6. Drive the vehicle out of the system.

## Components of Vertical Rotary Car Parking

TECHNOLOGICAL UNIVERSITY (MANDALAY)
DEPARTMENT OF MECHANICAL ENGINEERING

# “Vertical Rotary Car Parking”


## Fabrication Procedure

1. Cut and weld the hollow pipes into two U-shaped frames with supports.
2. Cut the 1-inch round bars into 24-inch and 28-inch shafts.
3. Drill four flat bars and weld them to the spur gears.
4. Install four ball bearings, fit the gears to the shafts, and insert the shafts.
5. Bend and assemble the iron sheets to form the parking slots.
6. Modify the chains and attach the triangular iron-sheet supports to the shafts.
7. Fit the parking slots to the chains.
8. Install a 40-RPM speed-controlled motor and connect it to the bottom shaft using a belt.

Making Frame

Making Shaft

Making Gear

Modified Chain

Parking Slots

Install Motor

Install Pulley

Installation and Finishing